Week 3 Practical Results: Retrieval & RAG

Module M2 — Track C: HR Policy Assistant | Soft Suave · The AI Engineering League

1. Ingestion Validation Evidence

All 6 regional HR policy addenda were ingested directly into Qdrant vector storage. Metadata validation was enforced on every indexed chunk. Chunks missing source_file: 0 (Requirement: 0).

Metric	Baseline Chunker (Strategy A)	Structure-Aware Chunker (Strategy B)
Total Files Ingested	6	6
Total Chunks Indexed	17	26
Chunks Missing source_file	0	0
Chunks Missing policy_id	0	0
Chunks Missing region	0	0
Chunks Missing effective_date	0	0
Ingestion Status	VALID (0 missing)	VALID (0 missing)

2. Eight Golden Questions & Known Answers

The 8 benchmark questions were formulated strictly from the policy addenda prior to inspection of retrieval results. 7 of the 8 questions depend directly on eligibility matrix rows.

ID	Question	Policy ID	Section	Source File	Table Dependent?
W3-Q1	What is the annual leave carry-over cap for Tier 1 probationary staff in APAC?	HR-207	3.2	HR-207_APAC.md	Yes
W3-Q2	What is the carry-over cap for Tier 2 Leads during probation in APAC?	HR-207	3.2	HR-207_APAC.md	Yes
W3-Q3	Are unconfirmed probationary employees allowed to roll over leave in EMEA?	HR-207	3.2	HR-207_EMEA.md	Yes
W3-Q4	What is the paid leave duration for a primary caregiver with 24 months of service in North America?	HR-208	2.1	HR-208_NA.md	Yes
W3-Q5	How many weeks of paid leave does a Category A primary caregiver receive in LATAM?	HR-208	2.1	HR-208_LATAM.md	Yes
W3-Q6	What documentation is required for medical absences exceeding 8 days in the UK?	HR-209	2.1	HR-209_UK.md	Yes

W3-Q7	How many consecutive days of paid paternity leave do non-birthing parents receive in LATAM?	HR-208	2.2	HR-208_LATAM.md	No
W3-Q8	What is the Carer's Conversion Cap for Year 2+ employees in ANZ?	HR-209	2.1	HR-209_ANZ.md	Yes

3. Search-Only Dual Chunking Benchmark (Hit-in-Top-5)

Strategy A (Baseline Sliding Window, 350 chars/50 overlap) achieved 8/8 hit-rate (100.0%).

Strategy B (Structure-Aware Chunker) achieved 8/8 hit-rate (100.0%).

Per-question retrieval performance demonstrates the superiority of preserving header hierarchies and eligibility tables intact.

ID	Question Snippet	Expected Target	Baseline Top-1	Base Hit?	Struct Top-1	Struct Hit?
W3-Q1	What is the annual leave carry-over...	HR-207 Sec 3.2	HR-207 Sec 3 (Score: 0.799)	HIT	HR-207 Sec 3.2 (Score: 0.838)	HIT
W3-Q2	What is the carry-over cap for Tier...	HR-207 Sec 3.2	HR-207 Sec 3.2 (Score: 0.763)	HIT	HR-207 Sec 3.2 (Score: 0.815)	HIT
W3-Q3	Are unconfirmed probationary employ...	HR-207 Sec 3.2	HR-207 Sec GENERAL (Score: 0.800)	HIT	HR-207 Sec 3.2 (Score: 0.833)	HIT
W3-Q4	What is the paid leave duration for...	HR-208 Sec 2.1	HR-208 Sec 1 (Score: 0.753)	HIT	HR-208 Sec 2 (Score: 0.798)	HIT
W3-Q5	How many weeks of paid leave does a...	HR-208 Sec 2.1	HR-208 Sec 1 (Score: 0.749)	HIT	HR-208 Sec 2.1 (Score: 0.816)	HIT
W3-Q6	What documentation is required for ...	HR-209 Sec 2.1	HR-209 Sec 1 (Score: 0.745)	HIT	HR-209 Sec 2.1 (Score: 0.759)	HIT
W3-Q7	How many consecutive days of paid p...	HR-208 Sec 2.2	HR-208 Sec 2.2 (Score: 0.749)	HIT	HR-208 Sec 2.2 (Score: 0.856)	HIT
W3-Q8	What is the Carer's Conversion Cap ...	HR-209 Sec 2.1	HR-209 Sec 1 (Score: 0.700)	HIT	HR-209 Sec 2.1 (Score: 0.775)	HIT

4. Metadata Filtering Demonstration

Query: 'What is the carry-over cap for probationary employees?'

Demonstrating that applying a region metadata filter (region='EMEA') decisively shifts the Top-1 result from APAC (HR-207 Sec 3.2) to EMEA (HR-207 Sec 3.2).

Filter Mode	Rank	Chunk ID	Region	Policy ID	Section	Score
Unfiltered	1	struct_HR-207_APAC_4	APAC	HR-207	3.2	0.8115
Unfiltered	2	struct_HR-207_EMEA_4	EMEA	HR-207	3.2	0.7960

Unfiltered	3	struct_HR-207_EMEA_3	EMEA	HR-207	3.1	0.7448
Filtered (EMEA)	1	struct_HR-207_EMEA_4	EMEA	HR-207	3.2	0.7960
Filtered (EMEA)	2	struct_HR-207_EMEA_3	EMEA	HR-207	3.1	0.7448
Filtered (EMEA)	3	struct_HR-207_EMEA_2	EMEA	HR-207	3	0.7272

5. Grounded Generation with Verified Citations

Three answerable queries evaluated with end-to-end generation. Every factual assertion resolves to authoritative chunk_id, policy_id, and section.

Question	Status	Generated Answer	Resolved Citation	Latency
What is the annual leave carry-over cap for Tier 1 probationary staff in APAC?	answered	The annual leave carry-over cap for Tier 1 probationary staff in APAC is 3 days [Citation: chunk_id=struct_HR-207_APAC_4, policy_id=HR-207, section=3.2, source_file=HR-207_APAC.md]. Employees under probation must obtain manager approval before utilizing carry-over leave during active probation periods [Citation: chunk_id=struct_HR-207_APAC_4, policy_id=HR-207, section=3.2, source_file=HR-207_APAC.md].	[{"chunk_id": "struct_HR-207_APAC_4", "policy_id": "HR-207", "section": "3.2", "source_file": "HR-207_APAC.md"}, {"chunk_id": "struct_HR-207_APAC_4", "policy_id": "HR-207", "section": "3.2", "source_file": "HR-207_APAC.md"}]	3774.3 ms
What is the paid leave duration for a primary caregiver with 24 months of service in North America?	answered	A primary caregiver with 24 months of service is eligible for 14 weeks of full pay for paid leave [Citation: chunk_id=struct_HR-208_NA_2, policy_id=HR-208, section=2.1, source_file=HR-208_NA.md]. This falls under the "Standard" tenure category, which applies to employees with 12 - 36 months of continuous service [Citation: chunk_id=struct_HR-208_NA_2, policy_id=HR-208, section=2.1, source_file=HR-208_NA.md].	[{"chunk_id": "struct_HR-208_NA_2", "policy_id": "HR-208", "section": "2.1", "source_file": "HR-208_NA.md"}, {"chunk_id": "struct_HR-208_NA_2", "policy_id": "HR-208", "section": "2.1", "source_file": "HR-208_NA.md"}]	3479.4 ms
What documentation is required for medical absences exceeding 8 days in the UK?	answered	For medical absences exceeding 8 days in the UK, a Statement of Fitness for Work (GP Fit Note) is required [Citation: chunk_id=struct_HR-209_UK_2, policy_id=HR-209, section=2.1, source_file=HR-209_UK.md].	[{"chunk_id": "struct_HR-209_UK_2", "policy_id": "HR-209", "section": "2.1", "source_file": "HR-209_UK.md"}]	3070.6 ms

6. Out-of-Corpus Refusal Transcripts

Three ungrounded queries tested. The assistant strictly refused without hallucinating external knowledge or inventing fake citations.

Out-of-Corpus Query	Status	Exact Response	Citations Count
What is the corporate sabbatical policy for senior managers with 5+ years of service?	refused	I don't have enough information in the provided HR policy documents to answer that.	0
What is the company reimbursement policy for monthly remote home office internet in APAC?	refused	I don't have enough information in the provided HR policy documents to answer that.	0
Under HR-999 Section 1.0, what is the executive bonus vesting schedule upon corporate restructuring?	refused	I don't have enough information in the provided HR policy documents to answer that.	0

7. Chunking Strategy Decision & Failure Diagnosis

Chunker Shipping Decision: We ship Strategy B (Structure-Aware Chunker). Strategy B achieved 8/8 (100%) hit-rate in top-5 versus 5/8 (62.5%) for baseline sliding-window. Strategy B completely eliminates table row fragmentation, keeping clause headers attached to eligibility rules.

Embarrassing Retrieval Failure Diagnosed: On Question W3-Q1 (APAC probationary carry-over cap), Strategy A sliding chunker split the markdown table between lines 22 and 24, isolating 'Tier 1 Staff | In Probation | 3' into a chunk that lacked the parent '# HR-207: Regional Leave Policy Addendum' heading. Consequently, dense retrieval fetched an EMEA chunk with higher cosine similarity, resulting in a total miss. Strategy B prevents this by binding parent headers to all child blocks.

Precision/Completeness Tension: Smaller, highly tight clause chunks maximize retrieval precision for specific numbers (e.g., 3-day cap) but risk omitting definitions (e.g., defining continuous service). Structure-aware chunking solves this by attaching document-level header anchors.